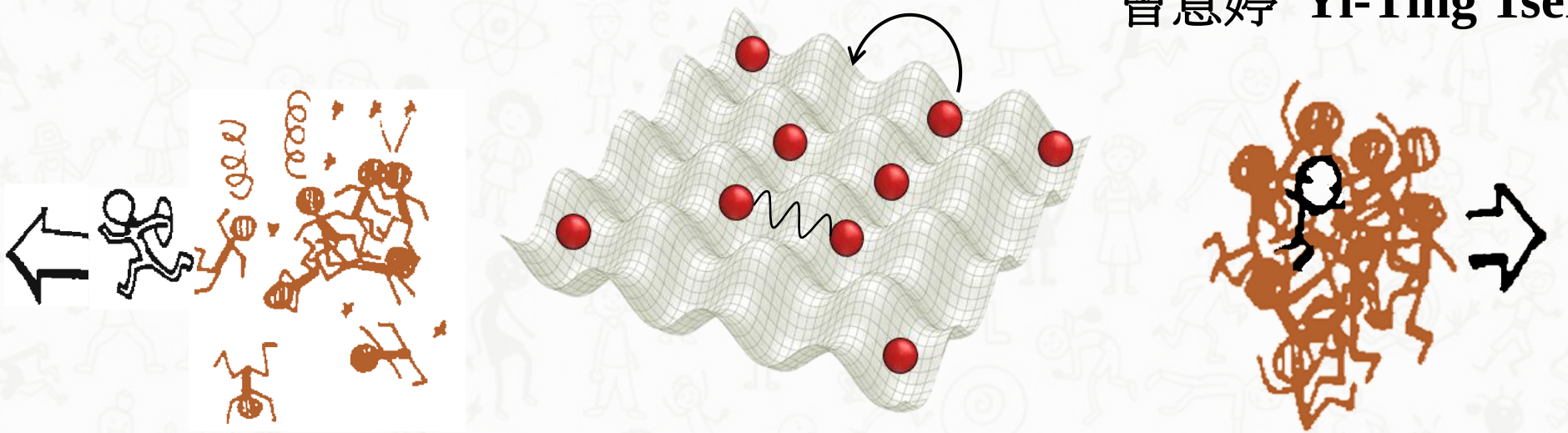
Correlated electrons in transition metal oxides: From CI to cluster DMFT

曾意婷 Yi-Ting Tseng



Bachelor

National Chung Hsing University ⚲ Taichung, Taiwan

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Chemical Physics of Solids: Physics of Correlated Matter

Ph.D. researcher

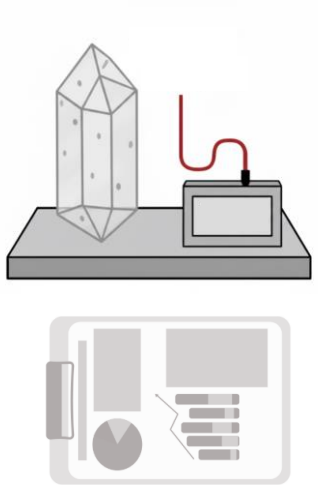
Universität Erlangen Nürnberg ⚲ Erlangen, Germany

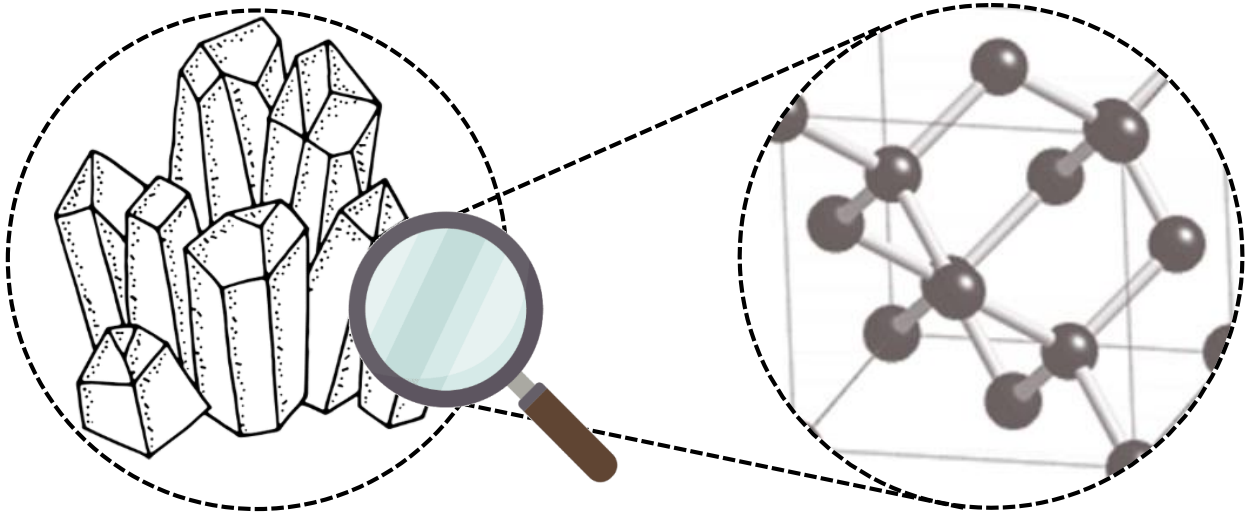
Institute of Theoretical Physics: AG Molecular Material



Correlated electrons in transition metal oxides: From CI to cluster DMFT

Solid State Physics





Macroscopic

Microscopic

materials like metals, semiconductors, …

how electrons behave within solids …



How to Solve the Schrödinger Eq.?

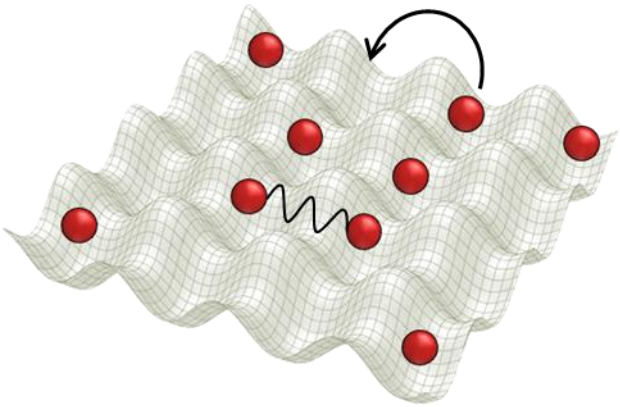
1927

Born-Oppenheimer Approximation



$$H_e = \underbrace{-\sum_i^{N_e} \frac{\hbar^2 \nabla_{\mathbf{r}_i}^2}{2m_e}}_{\text{kinetic}} \underbrace{-\sum_i^{N_e} \sum_l^{N_{\text{ion}}} \frac{Z_l e^2}{|\mathbf{r}_i - \mathbf{R}_l|}}_{\text{e-ion interaction}} + \underbrace{\sum_{j>i}^{N_e} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|}}_{\text{e-e interaction}}$$

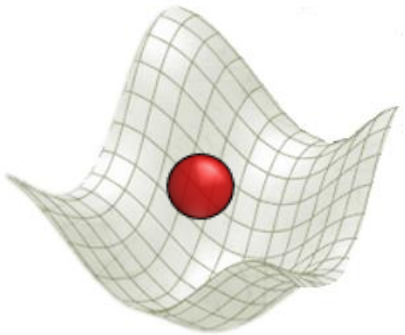




$$H\psi = E\psi$$ **: eigenvalue problem**

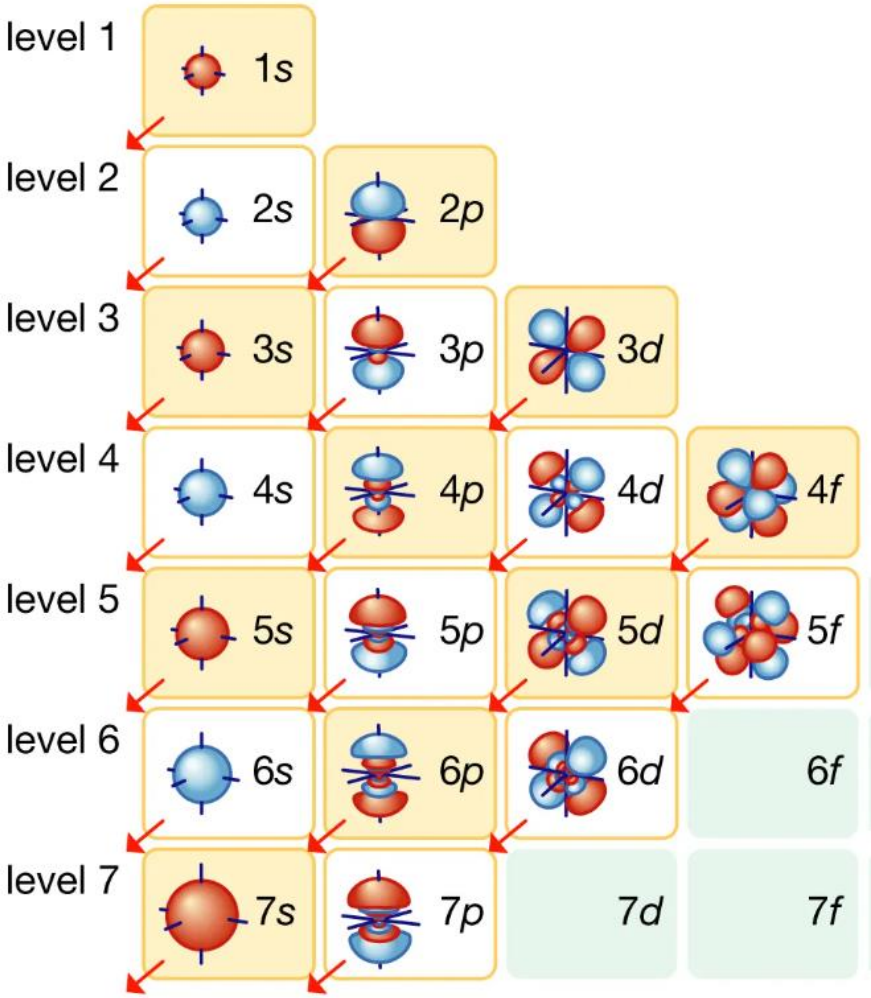
How to Solve the Schrödinger Eq.?

$$H_e = \underbrace{-\sum_i^{N_e} \frac{\hbar^2 \nabla^2_{\mathbf{r}_i}}{2m_e}}_{\text{kinetic}} \underbrace{-\sum_i^{N_e} \sum_l^{N_{\text{ion}}} \frac{Z_l e^2}{|\mathbf{r}_i - \mathbf{R}_l|}}_{\text{e-ion interaction}}$$



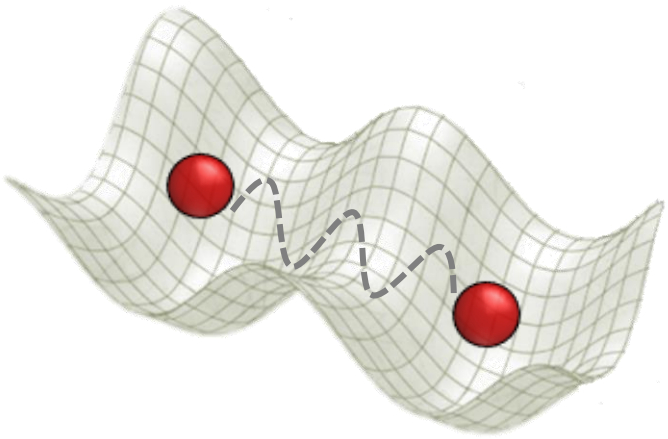
1 Hydrogen

electron configuration: s, p, d, f

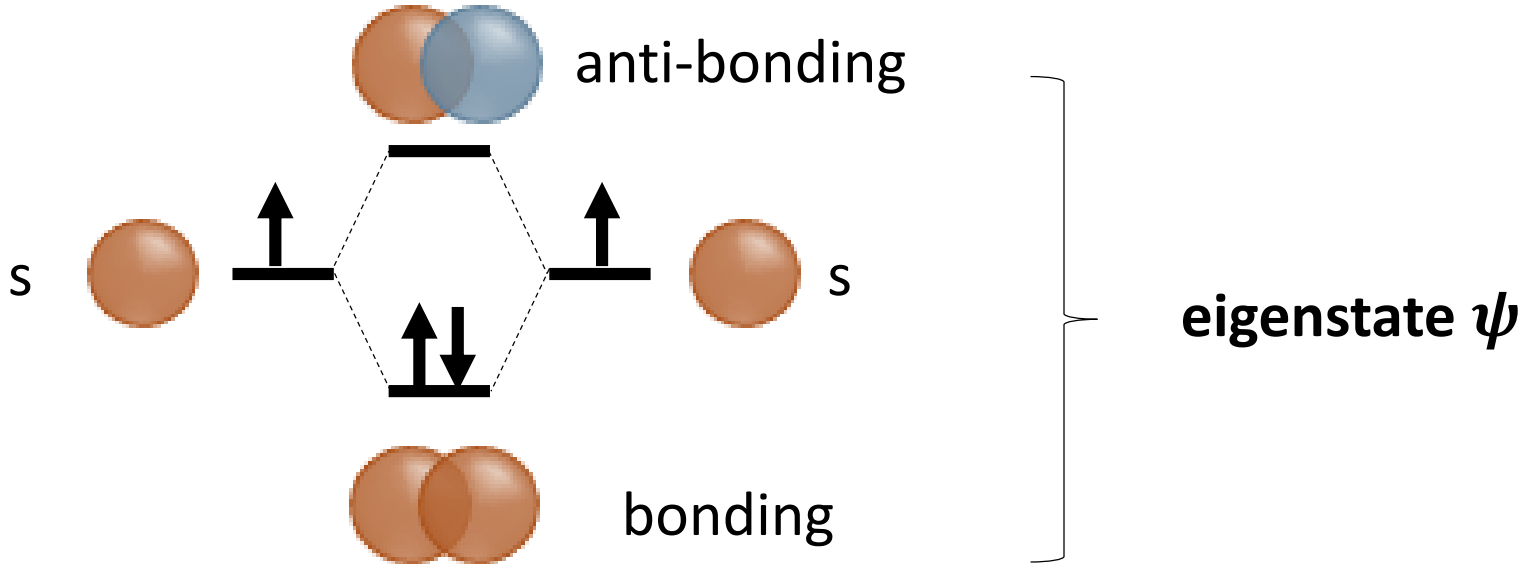


How to Solve the Schrödinger Eq.?

$$H_e = \underbrace{-\sum_i^{N_e} \frac{\hbar^2 \nabla^2_{\mathbf{r}_i}}{2m_e}}_{\text{kinetic}} \underbrace{-\sum_i^{N_e} \sum_l^{N_{\text{ion}}} \frac{Z_l e^2}{|\mathbf{r}_i - \mathbf{R}_l|}}_{\text{e-ion interaction}}$$

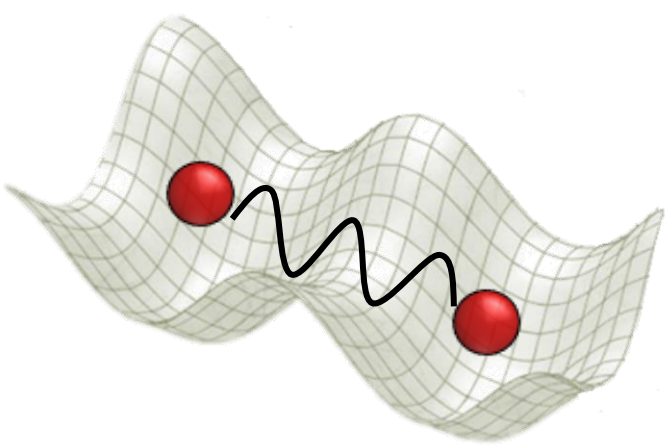


2 Hydrogen

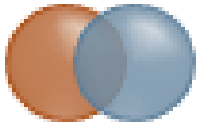


How to Solve the Schrödinger Eq.?

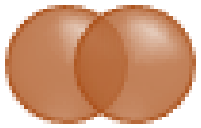
$$H_e = \underbrace{-\sum_i^{N_e} \frac{\hbar^2 \nabla^2_{\mathbf{r}_i}}{2m_e}}_{\text{kinetic}} \underbrace{-\sum_i^{N_e} \sum_l^{N_{\text{ion}}} \frac{Z_l e^2}{|\mathbf{r}_i - \mathbf{R}_l|}}_{\text{e-ion interaction}} + \underbrace{\sum_{j>i}^{N_e} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|}}_{\text{e-e interaction}}$$



2 Hydrogen



anti-bonding



bonding

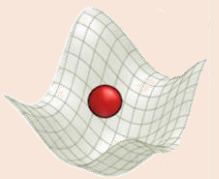
one of basis ϕ_i

Eigenstate ψ: linear combination of basis

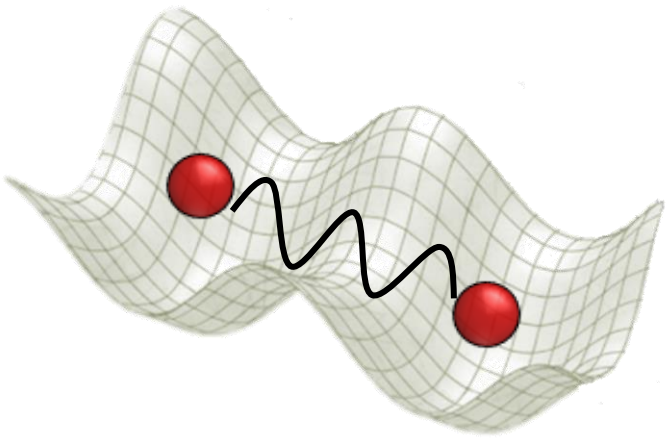
How to Solve the Schrödinger Eq.?

$$H_e = \underbrace{-\sum_i^{N_e} \frac{\hbar^2 \nabla^2_{\mathbf{r}_i}}{2m_e}}_{\text{kinetic}} \underbrace{-\sum_i^{N_e} \sum_l^{N_{\text{ion}}} \frac{Z_l e^2}{|\mathbf{r}_i - \mathbf{R}_l|}}_{\text{e-ion interaction}} + \underbrace{\sum_{j>i}^{N_e} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|}}_{\text{e-e interaction}}$$

# atoms	# spin-orbitals	# electrons	# states
1	2	1	$\binom{2}{1}$ 2
2	4	2	$\binom{4}{2}$ 6



↑ or ↓

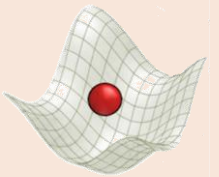


2 Hydrogen

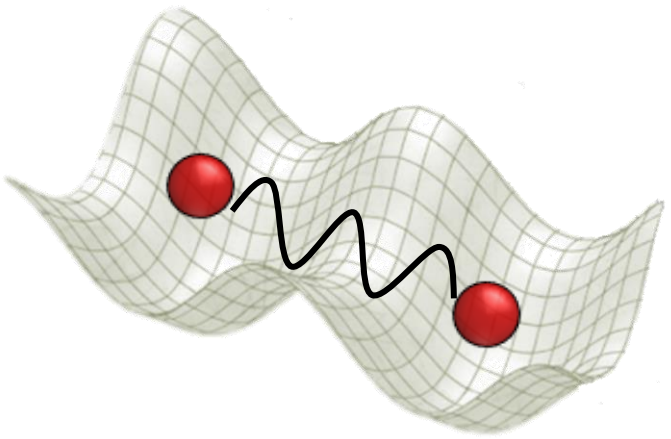
How to Solve the Schrödinger Eq.?

$$H_e = -\underbrace{\sum_i^{N_e} \frac{\hbar^2 \nabla^2_{\mathbf{r}_i}}{2m_e}}_{\text{kinetic}} - \underbrace{\sum_i^{N_e} \sum_l^{N_{\text{ion}}} \frac{Z_l e^2}{|\mathbf{r}_i - \mathbf{R}_l|}}_{\text{e-ion interaction}} + \underbrace{\sum_{j>i}^{N_e} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|}}_{\text{e-e interaction}}$$

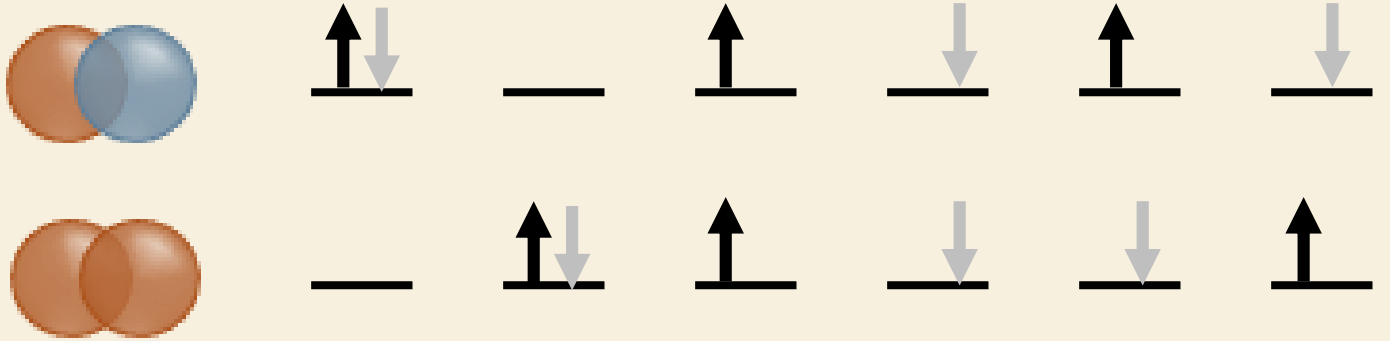
# atoms	# spin-orbitals	# electrons	# states
1	2	1	$\binom{2}{1}$ 2
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↑ or ↓



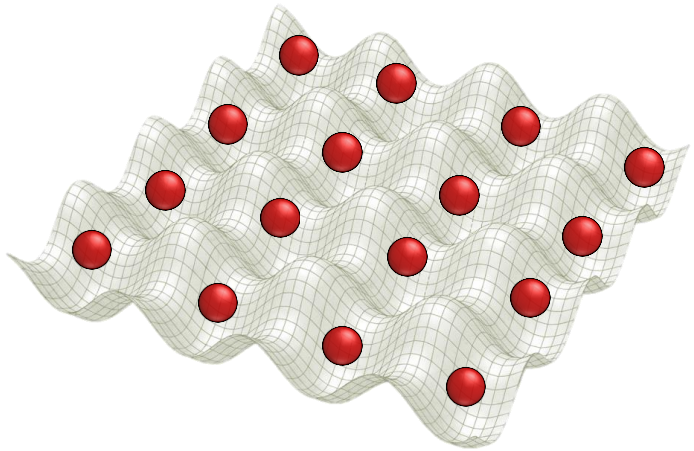
2 Hydrogen



Eigenstate: $\psi = \sum_{i=1}^{6} c_i \phi_i = c_1 \begin{bmatrix} \uparrow\downarrow \\ - \end{bmatrix} + c_2 \begin{bmatrix} - \\ \uparrow\downarrow \end{bmatrix} + c_3 \begin{bmatrix} \uparrow \\ \uparrow \end{bmatrix} + \cdots$

How to Solve the Schrödinger Eq.?

$$H_e = \underbrace{-\sum_{i}^{N_e} \frac{\hbar^2 \nabla^2_{\mathbf{r}_i}}{2m_e}}_{\text{kinetic}} \underbrace{-\sum_{i}^{N_e}\sum_{l}^{N_{\text{ion}}} \frac{Z_l e^2}{|\mathbf{r}_i - \mathbf{R}_l|}}_{\text{e-ion interaction}} + \underbrace{\sum_{j>i}^{N_e} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|}}_{\text{e-e interaction}}$$



16 Hydrogen

Configuration Interaction

$$\psi = \sum_{i=1}^{601080390} c_i \phi_i$$

# atoms	# spin-orbitals	# electrons	# states
1	**2**	**1**	$\binom{2}{1}$ **2**
2	**4**	**2**	$\binom{4}{2}$ **6**
4	8	4	$\binom{8}{4}$ 70
6	12	6	$\binom{12}{6}$ 924
8	16	8	$\binom{16}{8}$ 12 870
10	20	10	$\binom{20}{10}$ 184 756
12	24	12	$\binom{24}{12}$ 2 704 156
14	28	14	$\binom{28}{14}$ 40 116 600
16	**32**	**16**	$\binom{32}{16}$ **601 080 390**

Band Theory

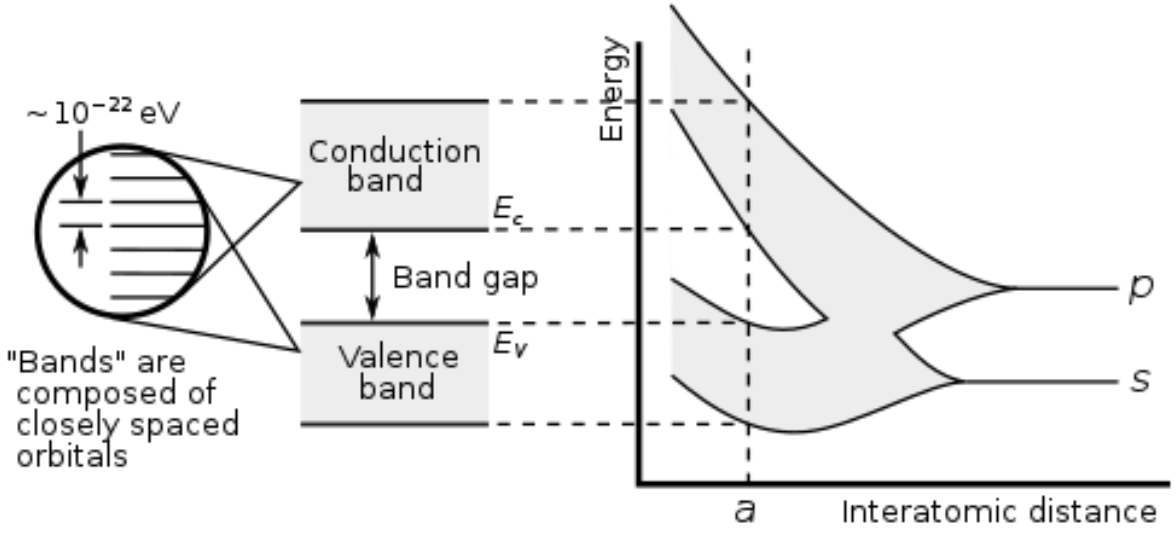
1928

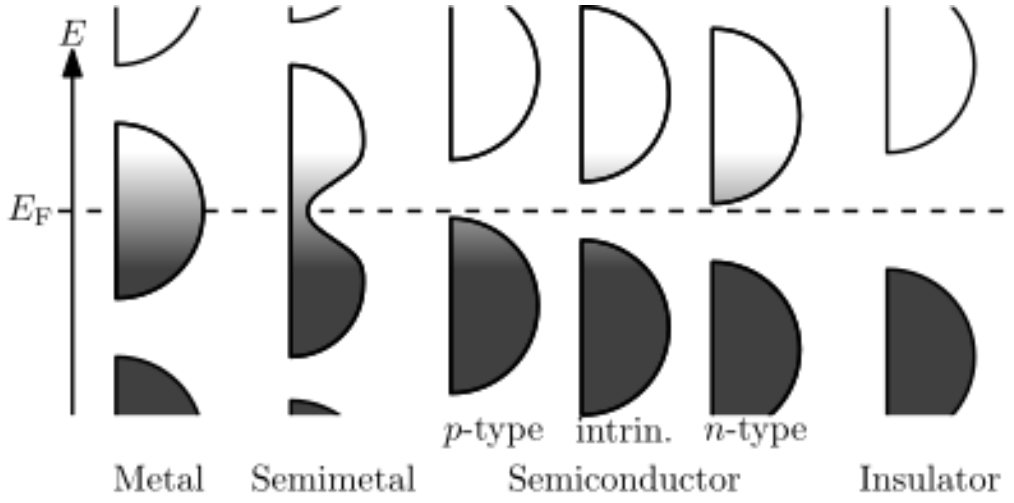
Bloch Theory

- periodic V(r): $\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r})$

wave functions overlap

valance band ; conduction band ; band gap





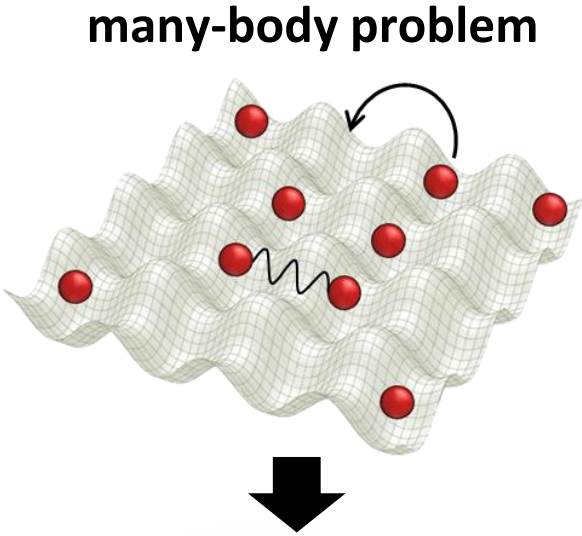
Band Theory & DFT

1964 -1965

Density Functional Theory (Hohenberg, Kohn, Sham)

- electron density functional $\rho(r) = \sum_i |\psi_i|^2$
- ground-state: min energy functional E[ρ]
- spatial coordinates: 3N → 3 (N-electron)

⇒ One Particle Approximation



$\mathrm{H}_{KS}\psi_k = E_k\psi_k$: Kohn Sham equation

$$E[\rho] = \underbrace{T_s[\rho]}_{\text{kinetic}} + \underbrace{\int \rho(\mathbf{r}) V_{\text{ext}}(\mathbf{r}) d\mathbf{r}}_{\text{e-ion interaction}} + \Bigg(\underbrace{\frac{1}{2} \iint \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} d\mathbf{r} d\mathbf{r}'}_{\text{e-e interaction}} + \underbrace{E_{\text{XC}}[\rho]}_{\text{exchange-correlation}} \Bigg)$$

$$\Bigg(-\frac{\hbar^2}{2m_e}\nabla^2 + \underbrace{V_{\text{ext}}(\mathbf{r}) + \int \frac{\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} d\mathbf{r}' + V_{\text{XC}}(\mathbf{r})}_{V_{\text{eff}}(\mathbf{r})} \Bigg) \psi_k(\mathbf{r}) = \epsilon_k \psi_k(\mathbf{r})$$

Fail of DFT

1964 -1965

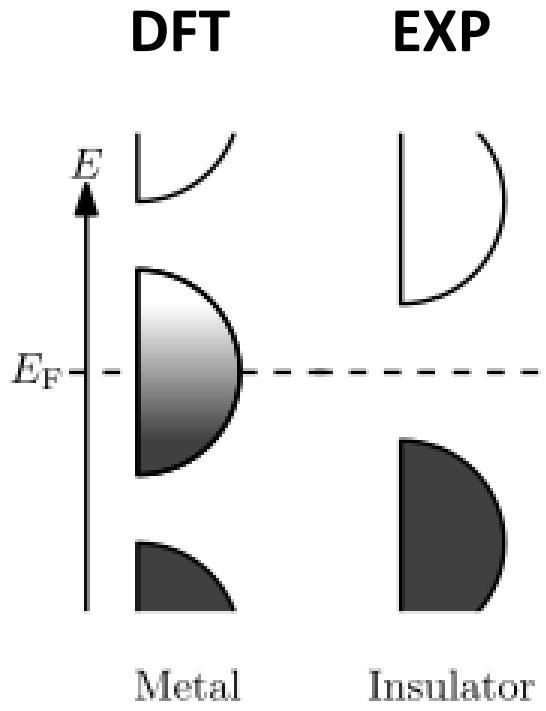
Density Functional Theory (Hohenberg, Kohn, Sham)

✓ **Metal, Semiconductor, Insulator**

✗ **NiO (Ni^{2+}, $3d^8$): partial filled → metal**

$$\psi = \sum_i c_i \phi_i \text{ correlated state } \rightarrow E_{XC}[\rho(r)]$$

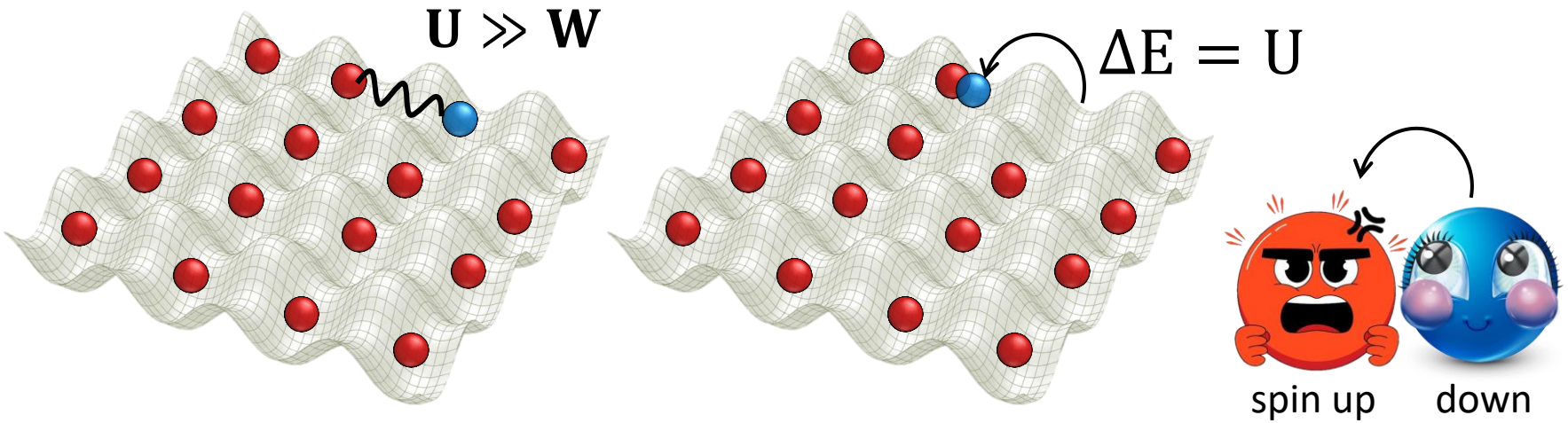
⇒ DFT underestimates **strong electronic correlations**



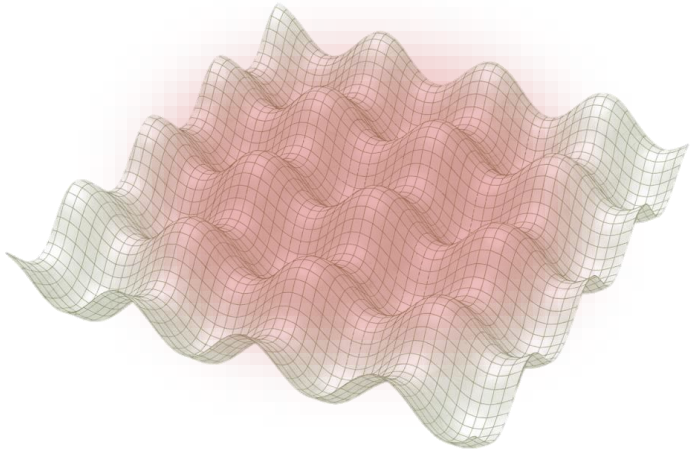
1937

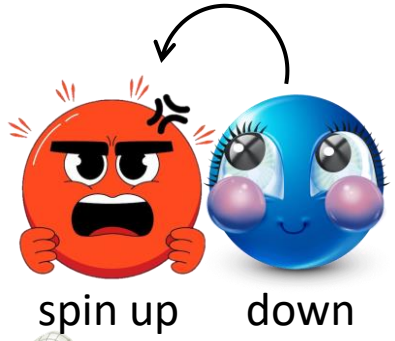
Mott Insulator

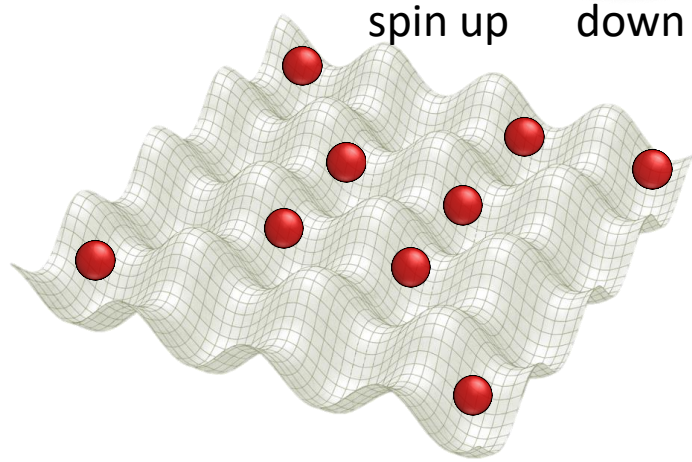
- **half-filled**



Itinerant and Localized Limit







Itinerant

$\mathbf{U = 0}$

Metals

Freely move

Bloch Wave

$\psi_k(r) = e^{ik \cdot r} u_k(r)$

Localized

$\mathbf{U \gg W}$

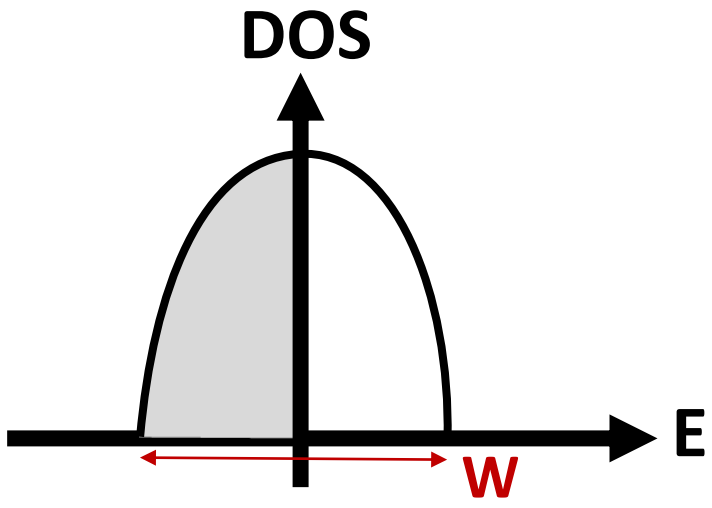
Mott insulator

confined

Atomic Configuration

s, p, d, f, …

Itinerant and Localized Limit



Itinerant

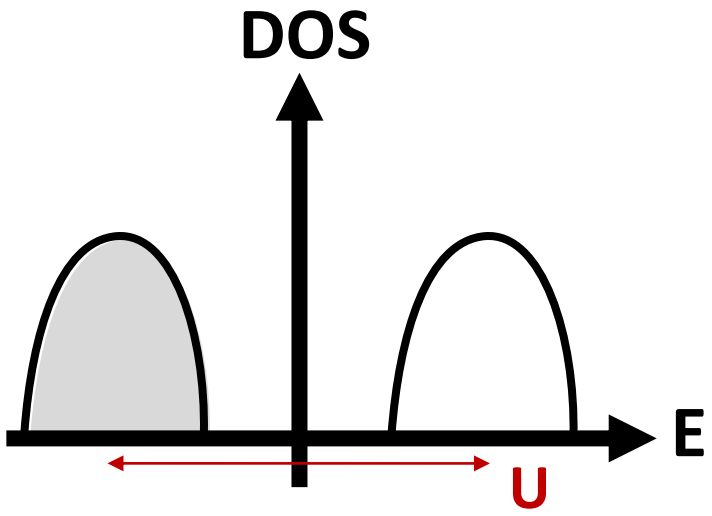
$\mathbf{U = 0}$

Metals

Freely move

Bloch Wave

$\psi_k(r) = e^{ik \cdot r} u_k(r)$



Localized

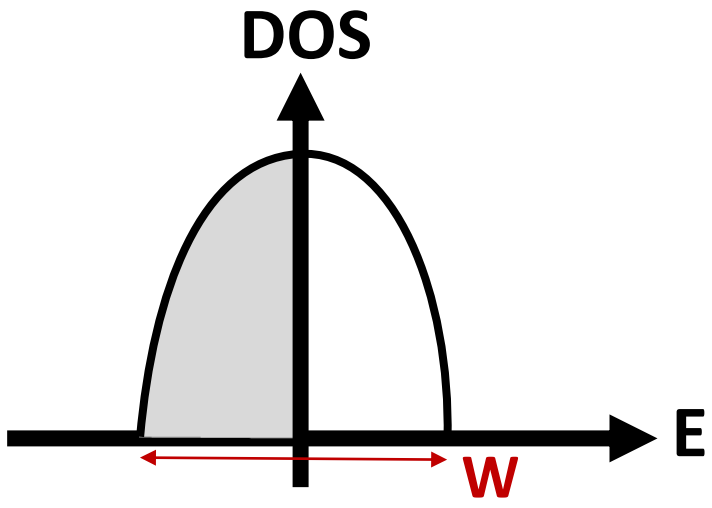
$\mathbf{U \gg W}$

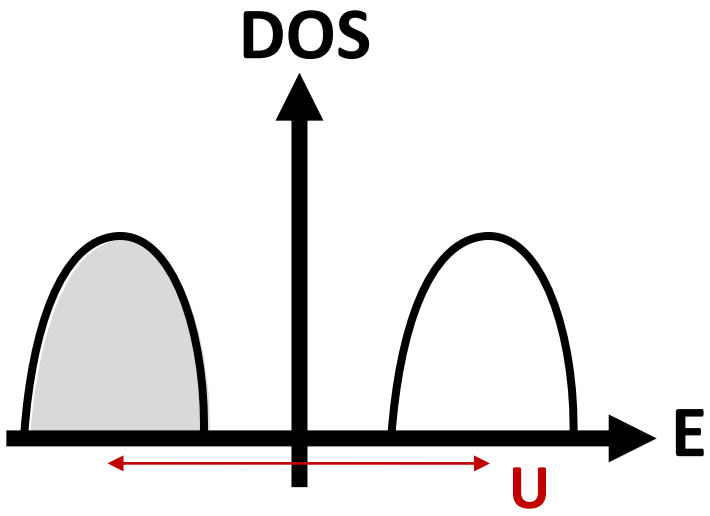
Mott insulator

confined

Atomic Configuration
s, p, d, f, …

Itinerant and Localized Limit





Itinerant

Localized

$\mathbf{U \approx W}$

$\mathbf{U = 0}$ ⟷ $\mathbf{U \gg W}$

Metals

Mott insulator

Freely move

confined

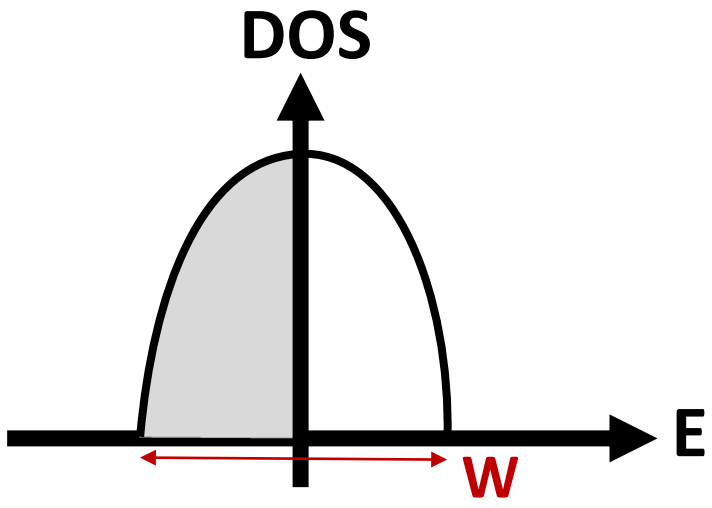
Bloch Wave

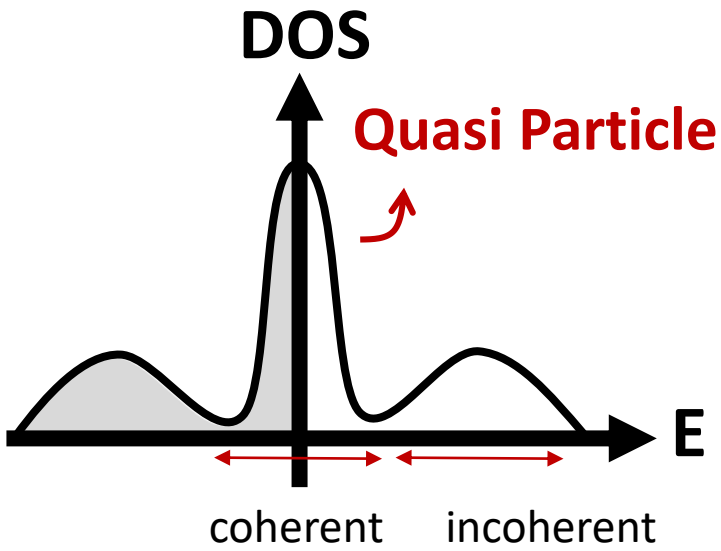
$\psi_k(r) = e^{ik \cdot r} u_k(r)$

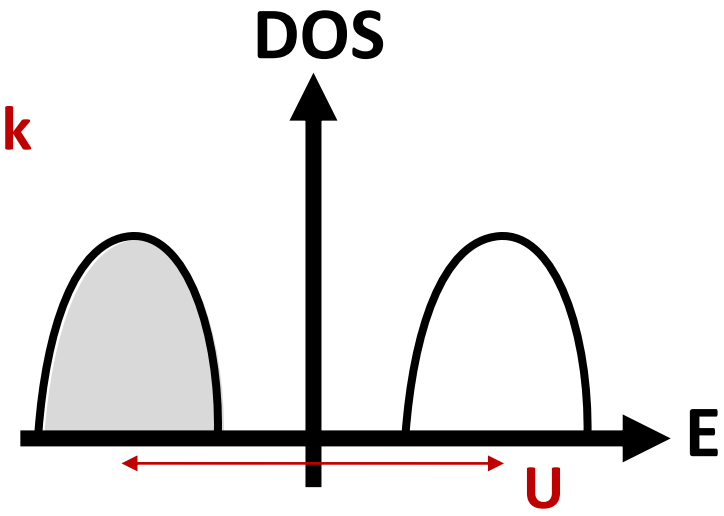
Atomic Configuration

s, p, d, f, …

Itinerant and Localized Limit







Itinerant

$\mathbf{U = 0}$

Metals

Freely move

Bloch Wave

$\psi_k(r) = e^{ik \cdot r} u_k(r)$

Strongly Correlated

Localized

$\mathbf{U \gg W}$

Mott insulator

confined

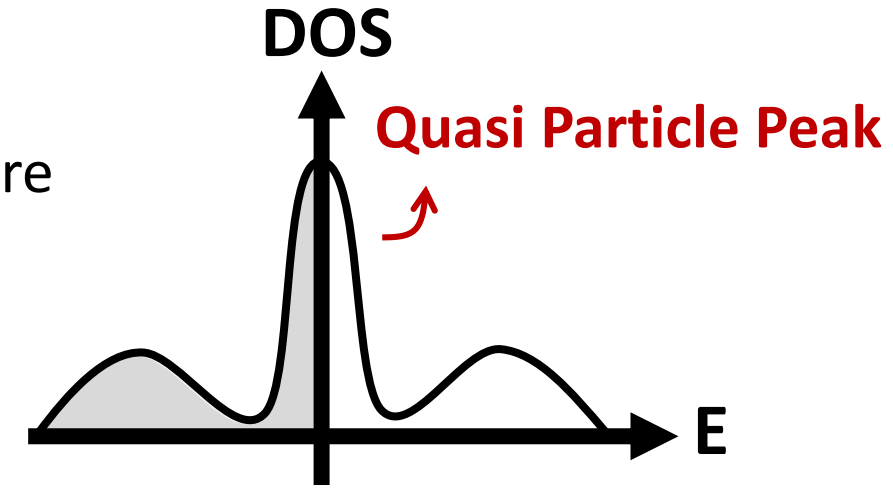
Atomic Configuration
s, p, d, f, …

Fermi Liquid Theory

1957 **Lev Davidovich Landau** refined band theory picture

- Quasiparticle: long-lived excitations
- collective behavior

⇒ "dressed" fermion, ex: $m^*/m_e > 1$



$$G^0(k,\omega) = [\omega + \mu - \varepsilon_k]^{-1}$$

$$G(k,\omega) = [\omega + \mu - \varepsilon_k - \boldsymbol{\Sigma(k,\omega)}]^{-1}$$

$\mathrm{Re}\,\Sigma(k,\omega)$: $\varepsilon_k \rightarrow \tilde{\varepsilon}_k$

$\mathrm{Im}\Sigma(k,\omega)$: $\tau^{-1} \propto a\omega^2 + bT^2$



free electron



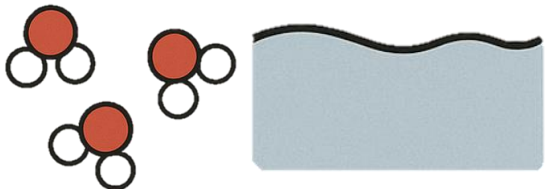
electron + cloud

More is Different

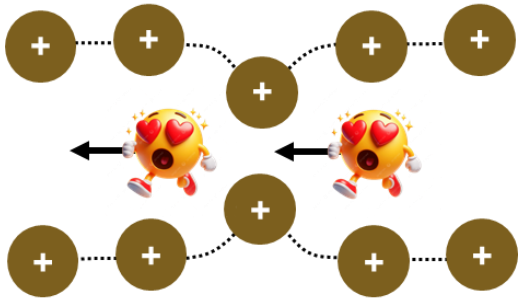
1972 **P. W. Anderson**

- **Emergence**
- **collective excitations → New phases**

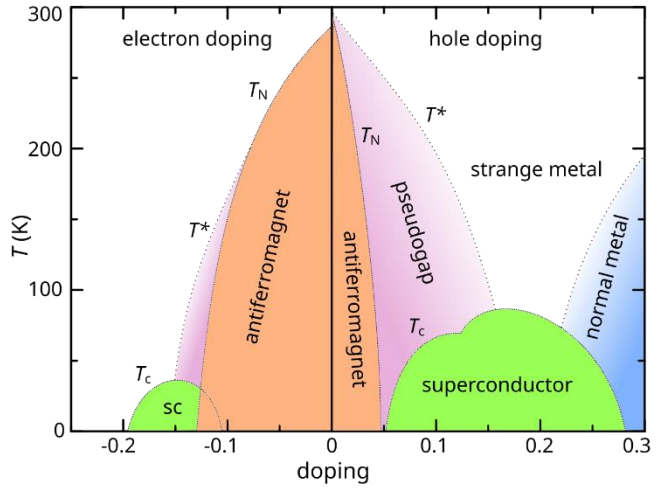
"The whole is not greater than the sum of its parts, it is different"



Liquid



Cooper Pair (1956)



Cuprate Phase Diagram (1986)



murmuration

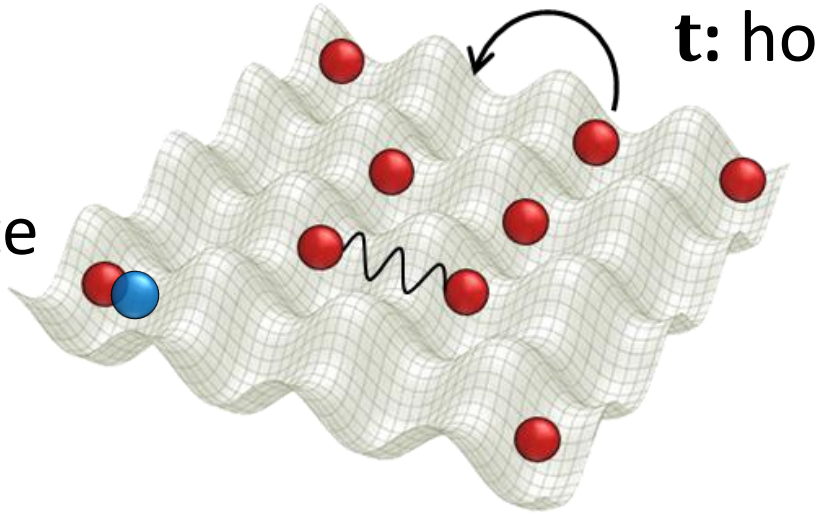
Effective Model

1963 **Hubbard Model (John Hubbard)**

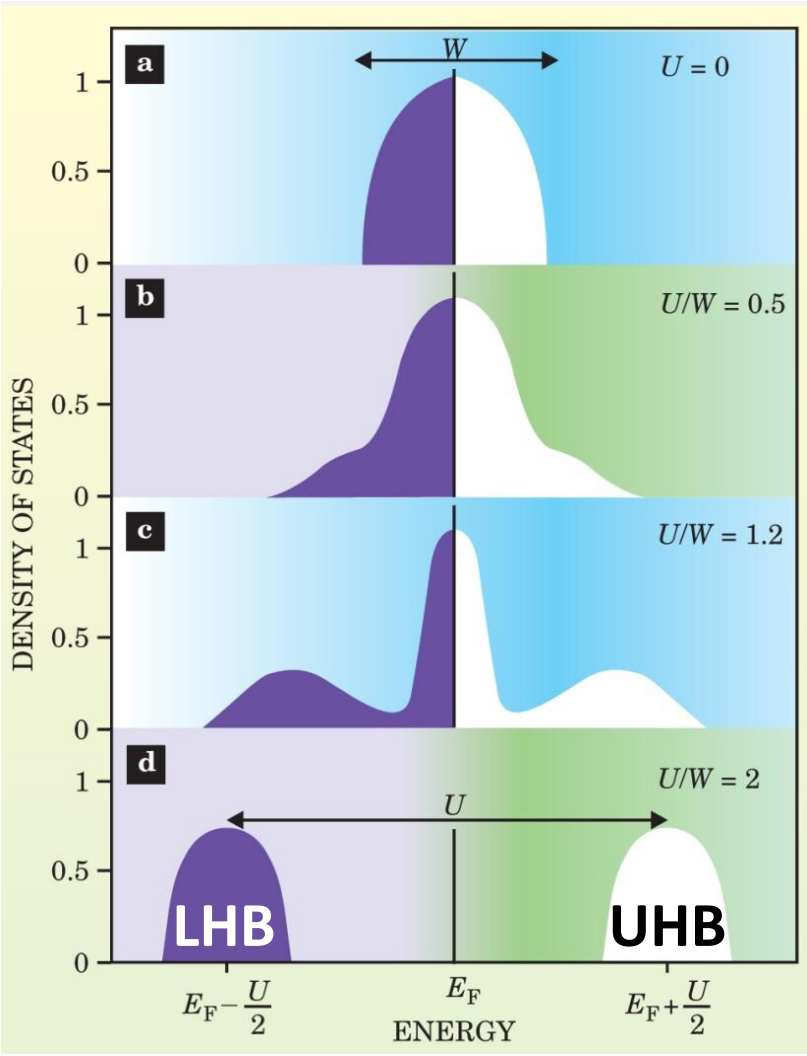
$$H = -t \sum_{\langle i,j \rangle, \sigma} \left(c^{\dagger}_{i\sigma} c_{j\sigma} + \text{h.c.} \right) + U \sum_{i} n_{i\uparrow} n_{i\downarrow}$$

t: hopping

U: onsite



Metal-Insulator Transition

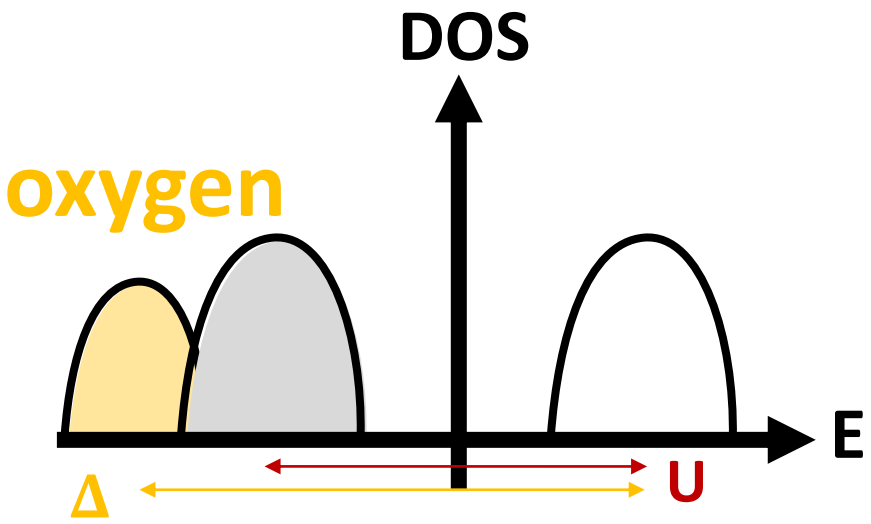


Effective Model

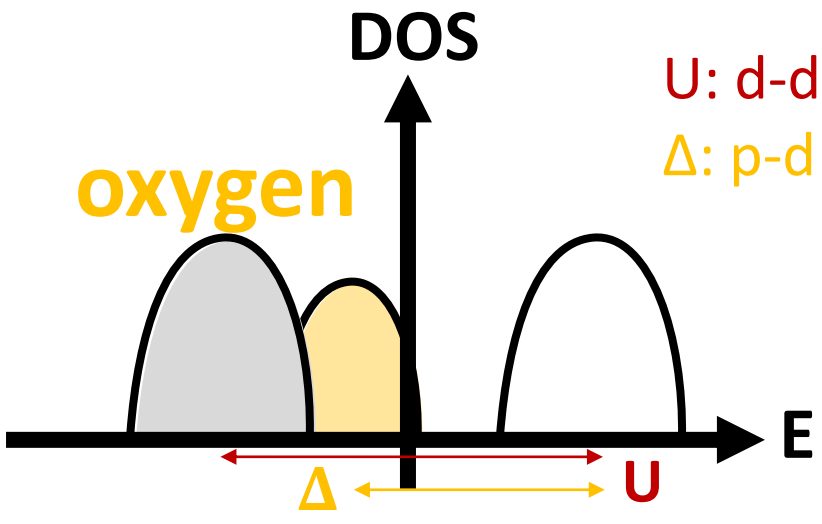
1963 **Hubbard Model (John Hubbard)**

$$H = -t \sum_{\langle i,j \rangle, \sigma} \left(c^{\dagger}_{i\sigma} c_{j\sigma} + \text{h.c.} \right) + U \sum_{i} n_{i\uparrow} n_{i\downarrow}$$

1985 **Zaanen-Sawatztky-Allen (ZSA) framework**



Mott Insulator



Charge Transfer Insulator

U: d-d excitation

Δ: p-d excitation

Effective Model

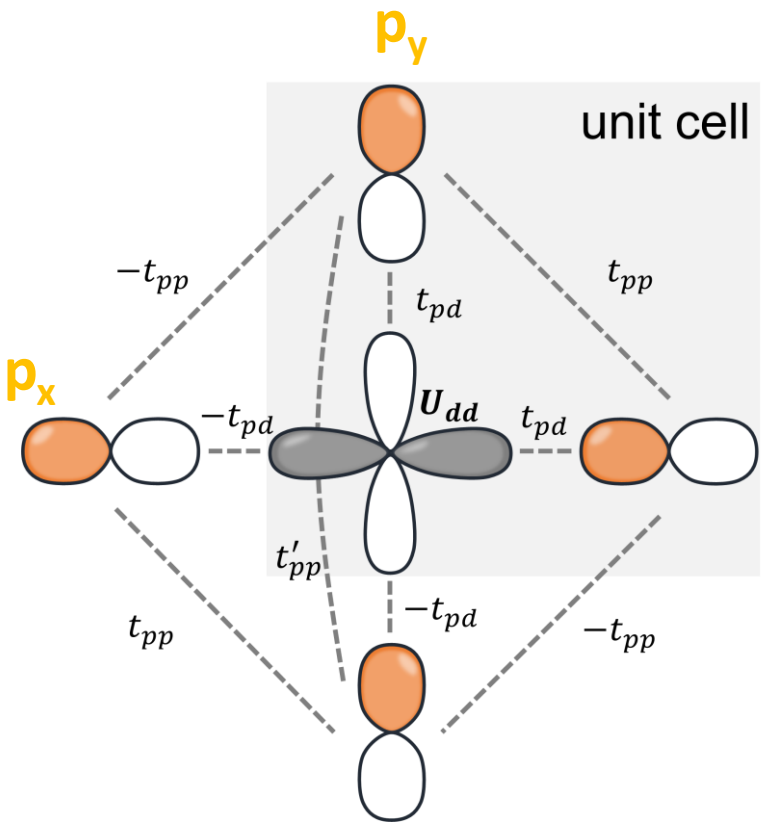
1963 **Hubbard Model (John Hubbard)**

$$H = -t \sum_{\langle i,j \rangle, \sigma} \left(c^{\dagger}_{i\sigma} c_{j\sigma} + \text{h.c.} \right) + U \sum_{i} n_{i\uparrow} n_{i\downarrow}$$

1985 **Zaanen-Sawatztky-Allen (ZSA) framework**

1987 **Emery Model (Victor J. Emery)**

$$H_{dp}(\mathbf{k}) = H_0^{dp}(\mathbf{k}) + H_{int.}^{dp}(\mathbf{k})$$



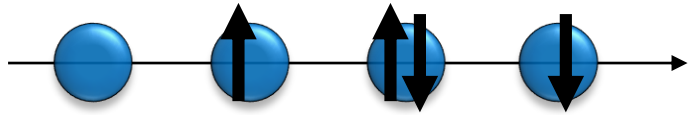
$$\mathrm{H}_0^{\mathrm{dp}}(\mathrm{k}) = \begin{array}{c} \begin{matrix} \quad\mathbf{d} & \quad\quad\mathbf{p_x} & \quad\quad\mathbf{p_y} \end{matrix} \\ \begin{pmatrix} \varepsilon_d & \mathrm{t_{pd}}\left(1 - \mathrm{e}^{-\mathrm{ik_x}}\right) & \mathrm{t_{pd}}\left(1 - \mathrm{e}^{-\mathrm{ik_y}}\right) \\ \mathrm{t_{pd}}\left(1 - \mathrm{e}^{\mathrm{ik_x}}\right) & \varepsilon_\mathrm{p} + 2\mathrm{t'_{pp}}\cos \mathrm{k_x} & \mathrm{t_{pp}}\left(1 - \mathrm{e}^{\mathrm{ik_x}}\right)\left(1 - \mathrm{e}^{-\mathrm{ik_y}}\right) \\ \mathrm{t_{pd}}\left(1 - \mathrm{e}^{\mathrm{ik_y}}\right) & \mathrm{t_{pp}}\left(1 - \mathrm{e}^{-\mathrm{ik_x}}\right)\left(1 - \mathrm{e}^{\mathrm{ikk_y}}\right) & \varepsilon_\mathrm{p} + 2\mathrm{t'_{pp}}\cos \mathrm{k_y} \end{pmatrix} \end{array}$$

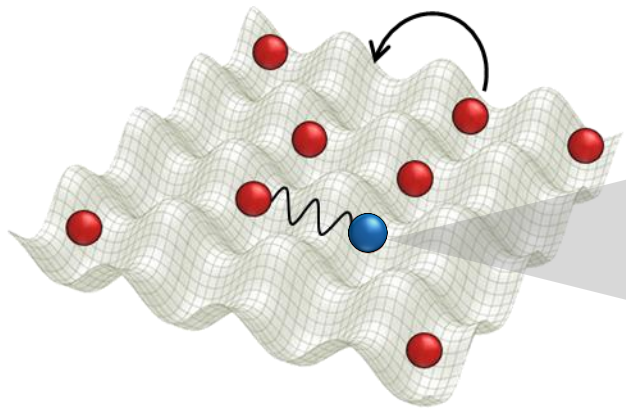
Dynamical Mean Field Theory

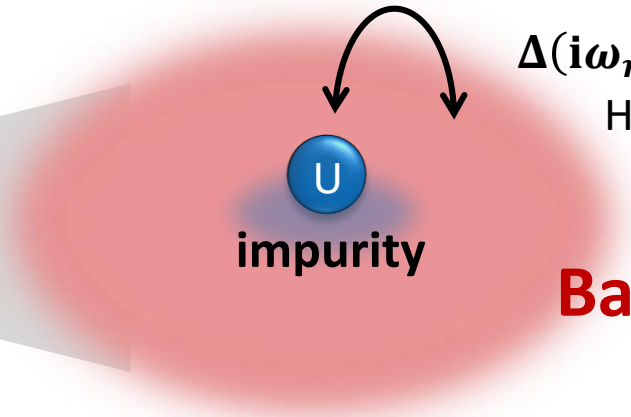
1989 **Walter Metzner and Dieter Vollhardt**

- map many body lattice problem into impurity + bath (exact: dim→ ∞)
- locality: $\Sigma(\mathrm{i}\omega_n, \mathrm{k}) \rightarrow \Sigma(\mathrm{i}\omega_n)$
- Dyson equation: self-consistency loop

$$G(i\omega_n, \mathbf{k}) = G_0(i\omega_n, \mathbf{k}) + G_0(i\omega_n, \mathbf{k})\Sigma(i\omega_n, \mathbf{k})G(i\omega_n, \mathbf{k})$$







$$\boldsymbol{\Delta(\mathrm{i}\omega_n) = \sum |V_p^{\sigma}|^2 / (i\omega_n - \varepsilon_p)}$$

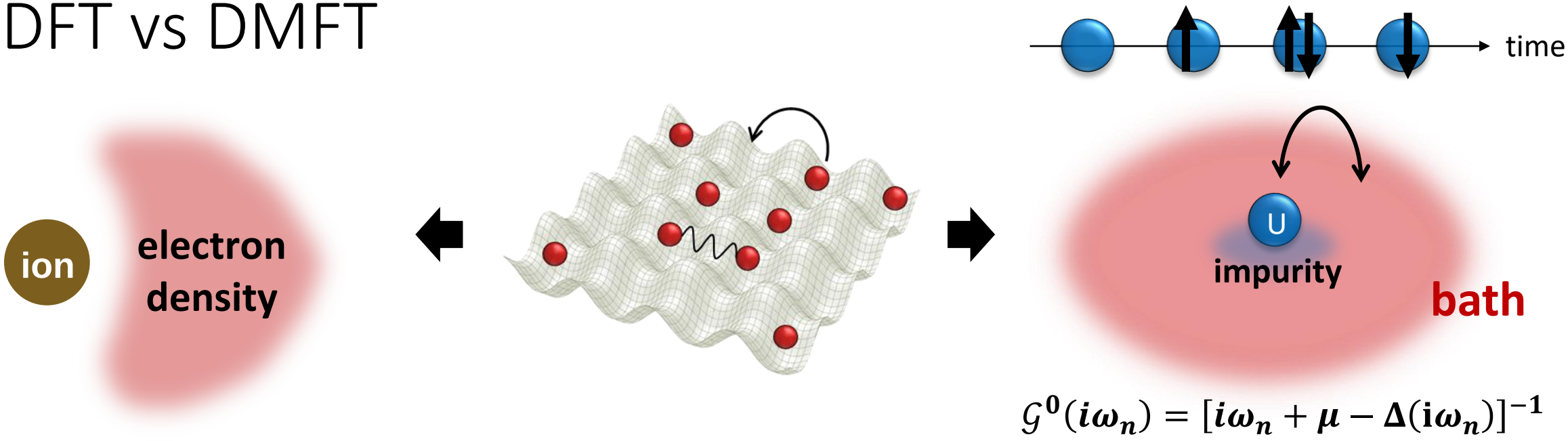
Hybridization function

Bath (ε_p & hopping V_p)

$$\mathcal{G}^0(i\omega_n) = [i\omega_n + \mu - \Delta(\mathrm{i}\omega_n)]^{-1}$$

$$\text{cf. } G^0(k, i\omega) = [i\omega_n + \mu - \varepsilon_k]^{-1}$$

DFT vs DMFT



$$\mathcal{G}^{0}(i\omega_n) = [i\omega_n + \mu - \Delta(i\omega_n)]^{-1}$$

Density Functional Theory

One particle approximation

Electrons in effective potential

Kohn-Sham potential $V_{eff}[\rho]$

Density $\rho(r) = \langle \psi^{\dagger}(r)\psi(r) \rangle$

Dynamical Mean Field Theory

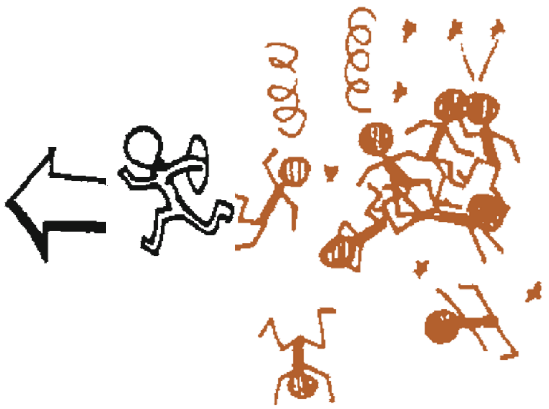
Many body approximation

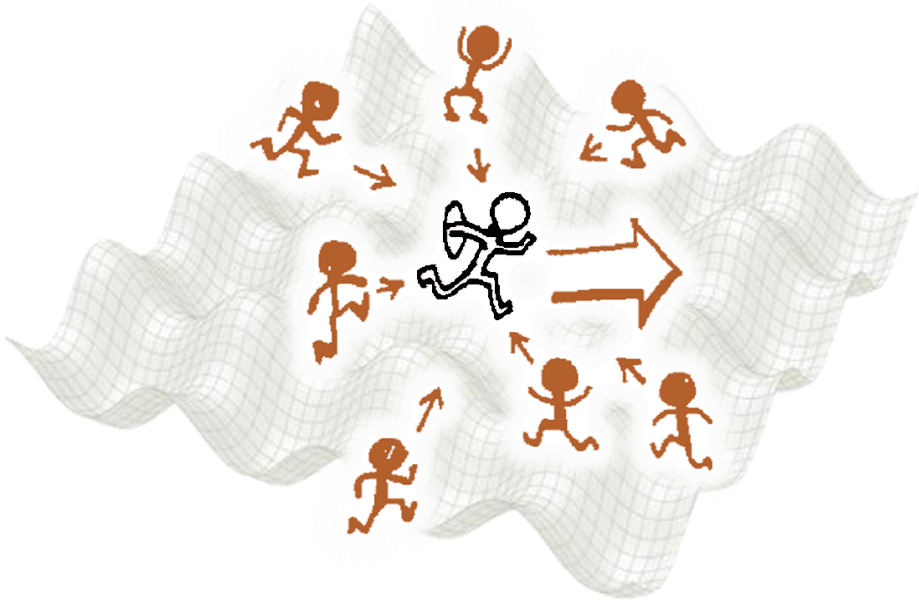
Impurity model hybridized with bath

Weiss field $\mathcal{G}^{0}(i\omega_n)$

Green Function $G(i\omega_n) = \langle c_i^{\dagger}(i\omega_n) c_j(i\omega_n) \rangle$

DFT vs DMFT







Density Functional Theory

One particle approximation

Electrons in effective potential

Kohn-Sham potential $V_{eff}[\rho]$

Density $\rho(r) = \langle \psi^{\dagger}(r)\psi(r) \rangle$

Dynamical Mean Field Theory

Many body approximation

Impurity model hybridized with bath

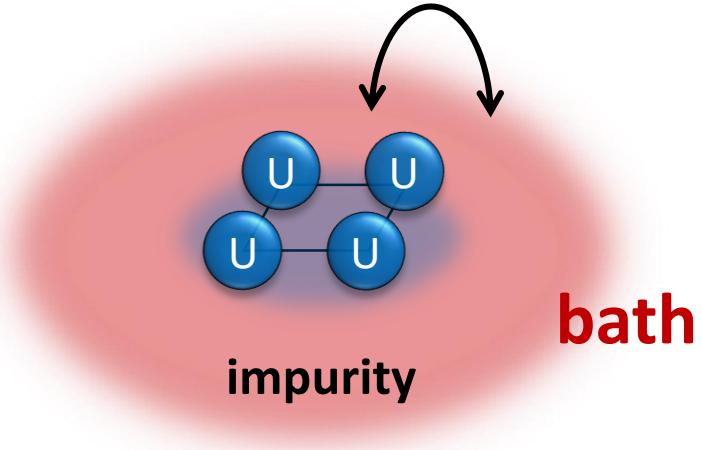
Weiss field $\mathcal{G}^{0}(i\omega_n)$

Green Function $G(i\omega_n) = \langle c_i^{\dagger}(i\omega_n)c_j(i\omega_n) \rangle$

DMFT Extension

cluster DMFT

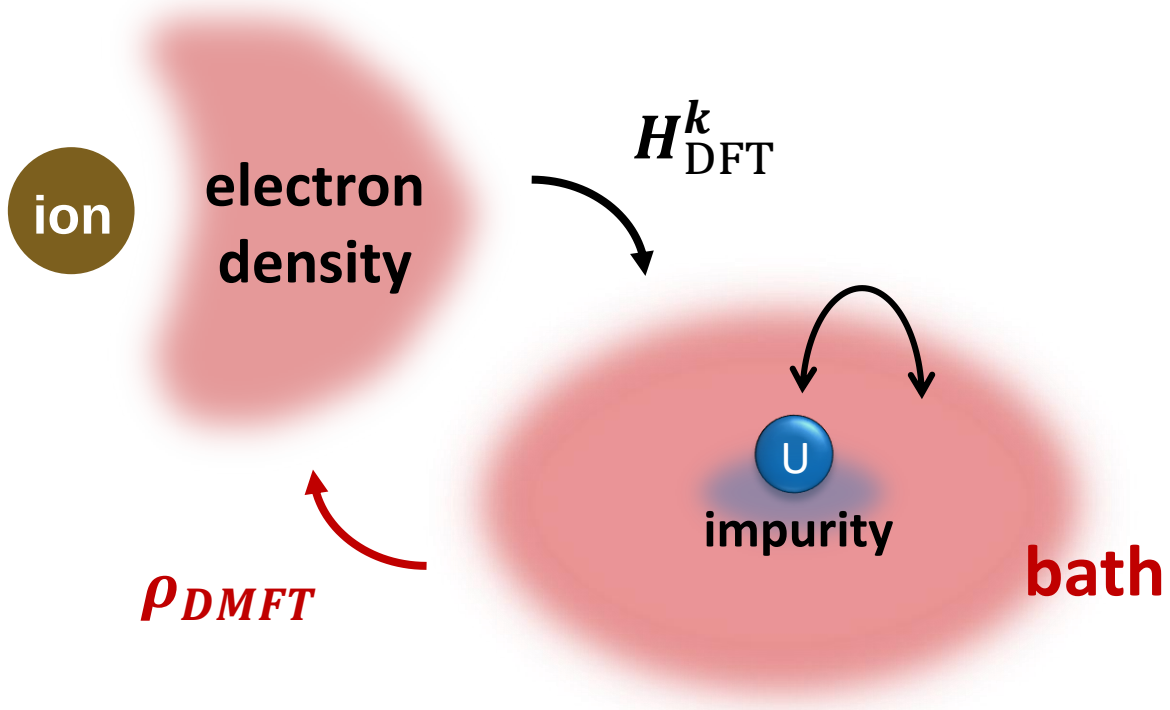
include short range correlation



$$\Sigma_{d_1 d_1} \rightarrow \begin{pmatrix} \Sigma_{d_1 d_1} & \boldsymbol{\Sigma_{d_1 d_2}} & \cdots \\ \boldsymbol{\Sigma_{d_2 d_1}} & \Sigma_{d_2 d_2} & \\ \vdots & & \ddots \end{pmatrix}$$

DFT+DMFT
charge self-consistency loop

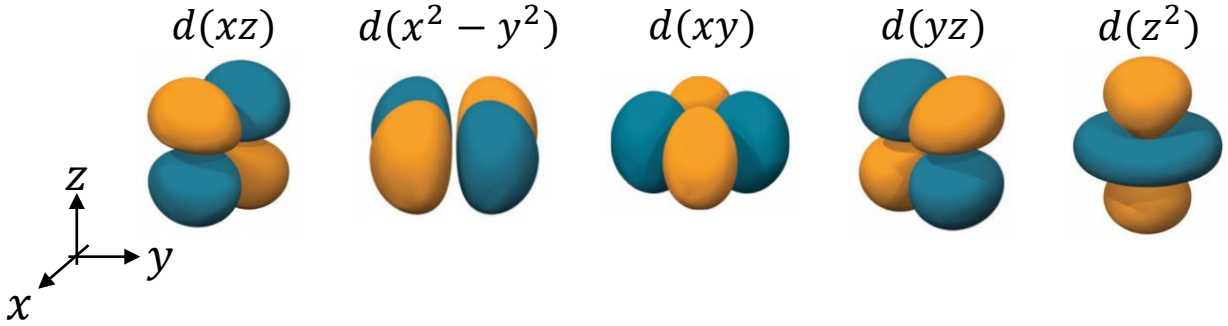
realistic band structure + local electronic correlations

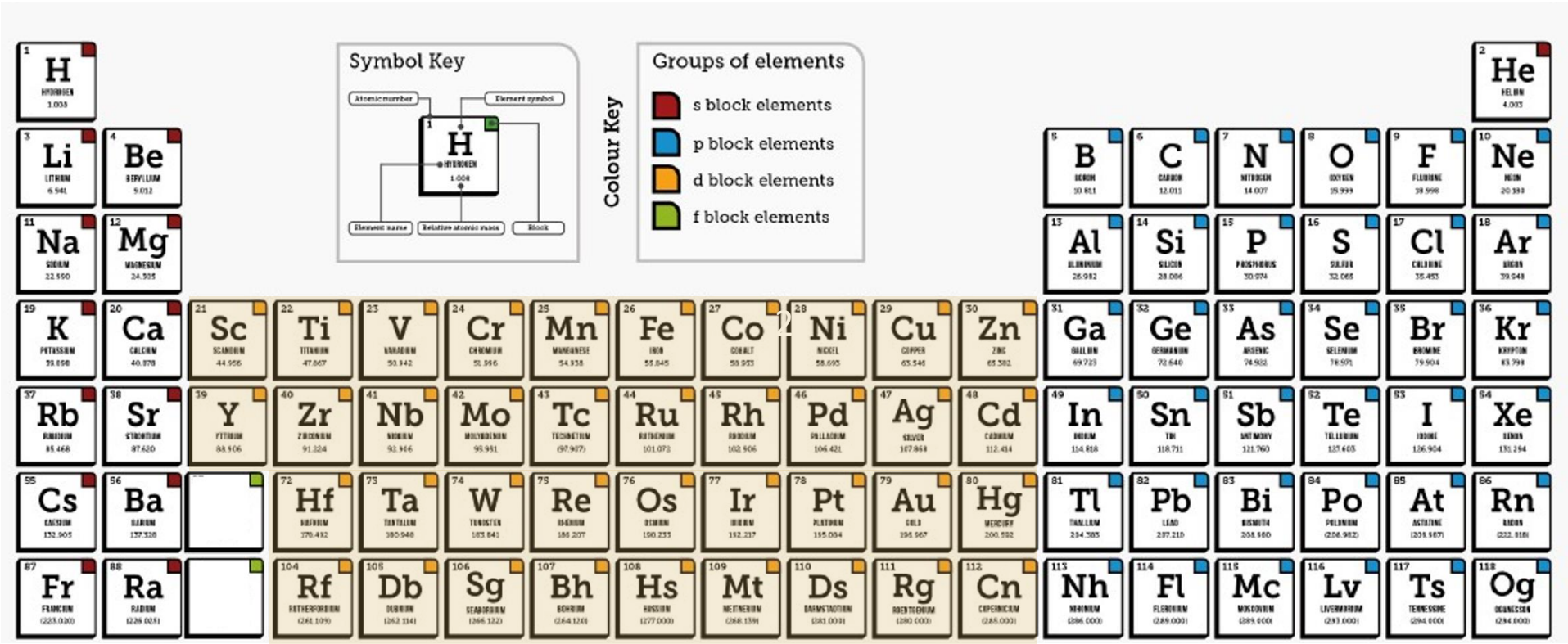


Transition Metal Oxides

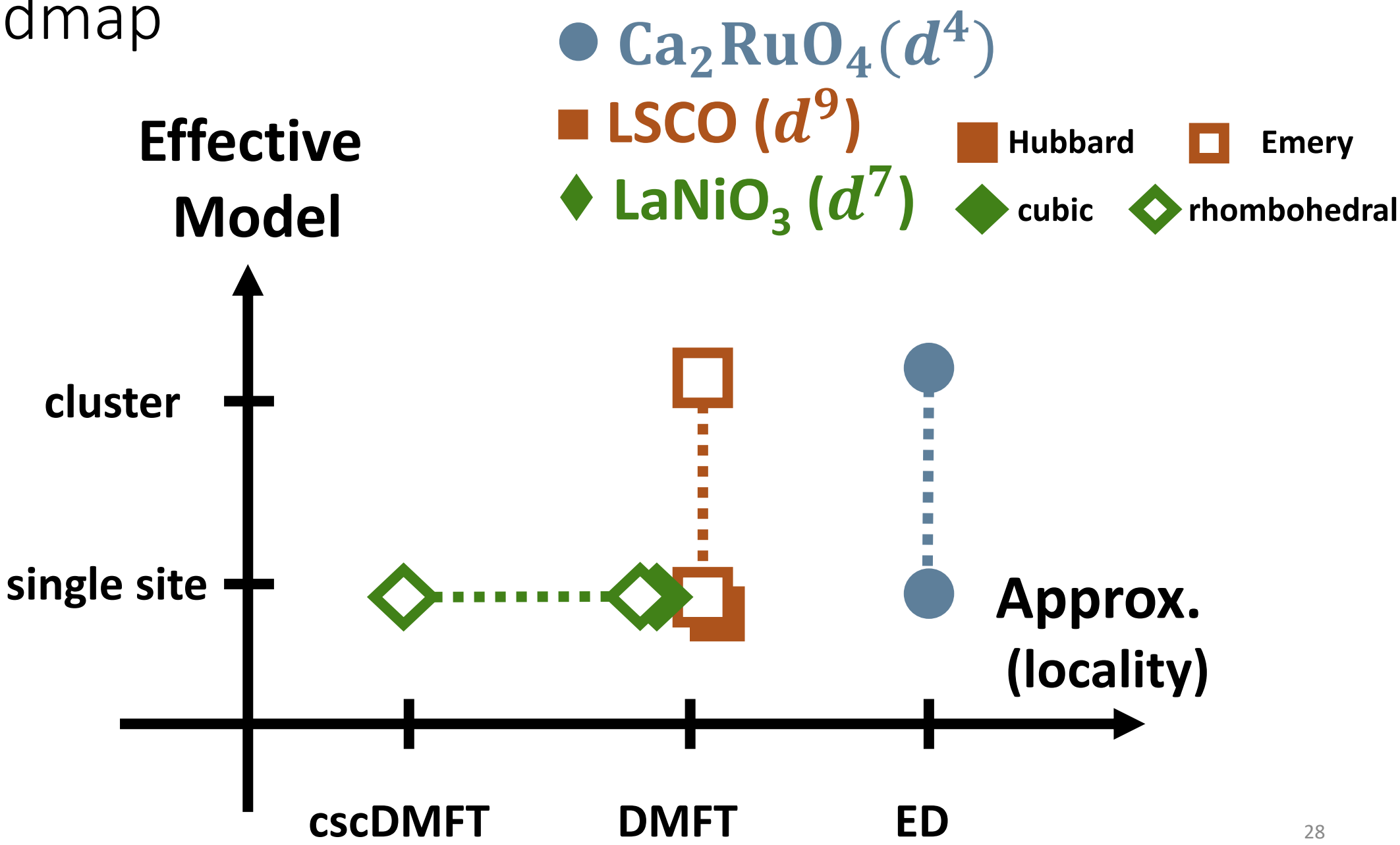
localized d-orbitals

- $U \gg W$ $\left(t = \int d^3r \, \phi_i^*(\mathbf{r}) \, \hat{H} \, \phi_{i+1}(\mathbf{r}) \right)$

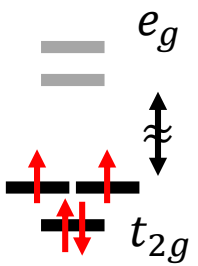


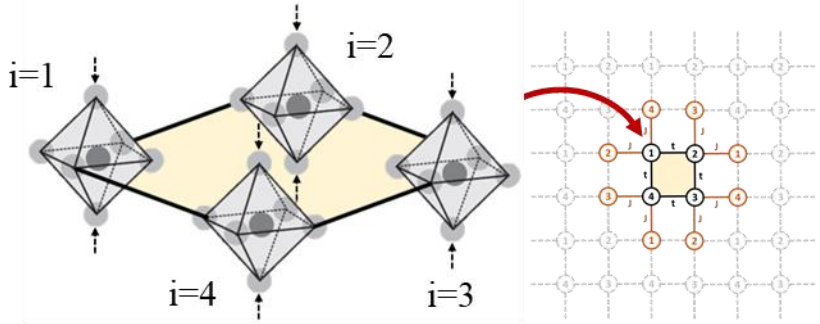


Roadmap


$Ca_2RuO_4(d^4)$
LSCO (d^9)
$LaNiO_3$ (d^7)
Hubbard
Emery
cubic
rhombohedral
Effective Model
cluster
single site
Approx. (locality)
cscDMFT
DMFT
ED

Ca_2RuO_4

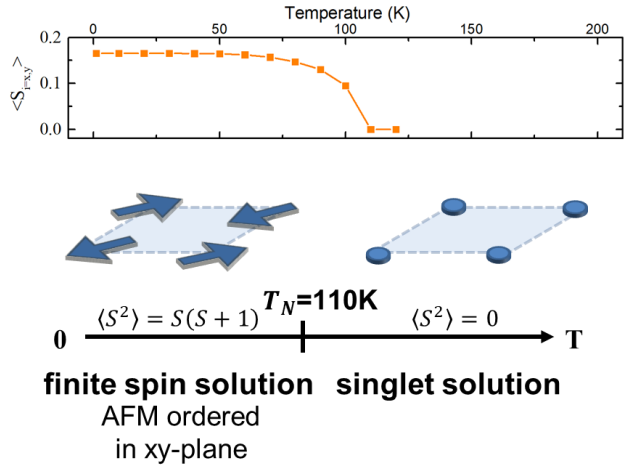




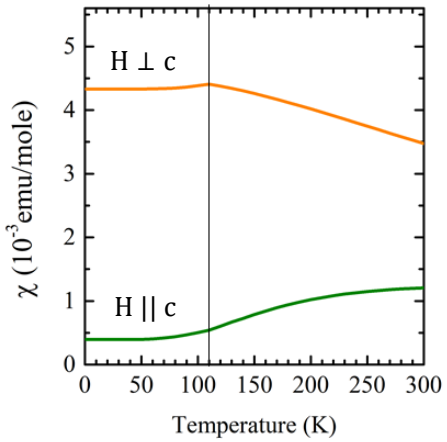
2×2 cluster t_{2g}-model + static MF + ED

- ✓ singlet - AFM transition ~ 110 K
- ✓ peak position
- × spectra weight transfer in PES

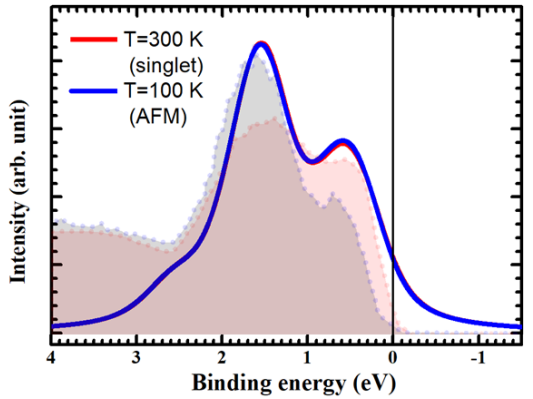
⇒ spin-polarons beyond static MF approach



spin moment



spin susceptibility



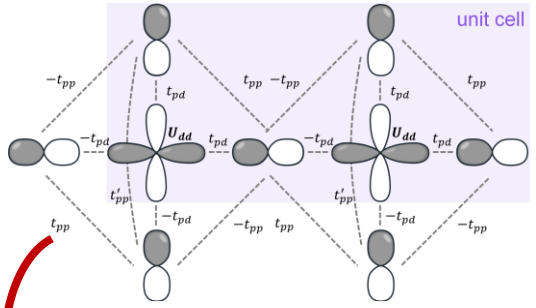
spectral function

Cuprate

singe site DMFT + Emery model

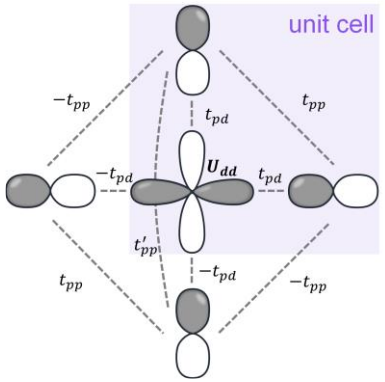
✓ $K \propto \chi_{dd}(\omega = 0, q = 0)$

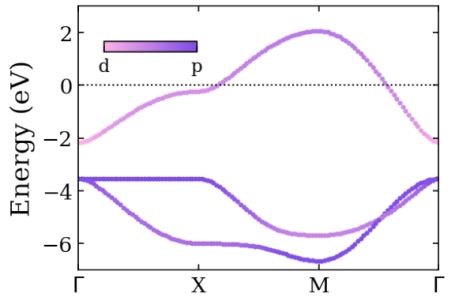
⇒ copper-oxygen correlation



$$\Sigma_{d_1 d_1} \rightarrow \begin{bmatrix} \Sigma_{d_1 d_1} & \Sigma_{d_1 d_2} \\ \Sigma_{d_2 d_1} & \Sigma_{d_2 d_2} \end{bmatrix}$$

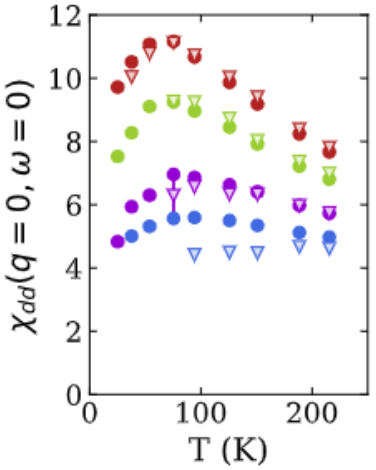
Emery

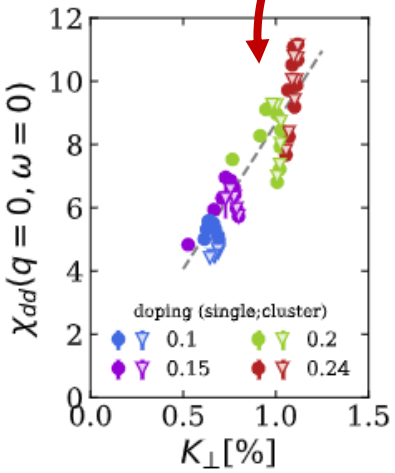




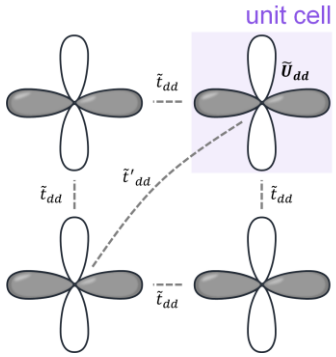
mixed k-dep.

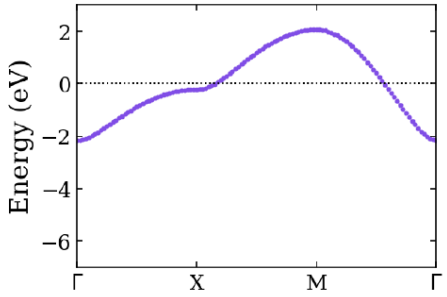
orbital character





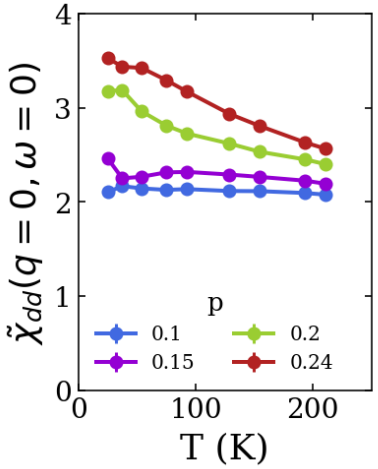
Hubbard

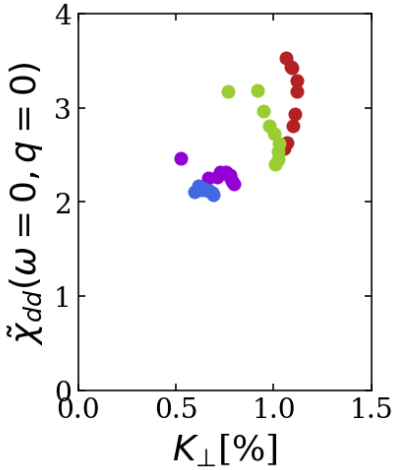




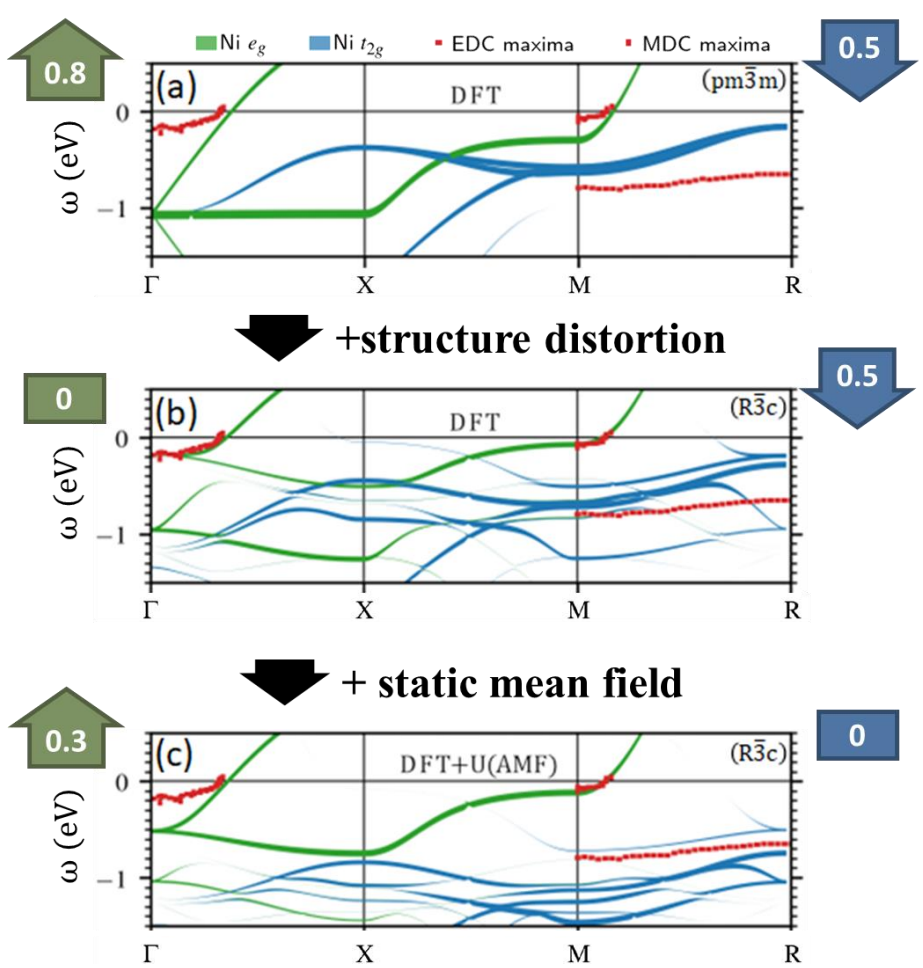
no orbital

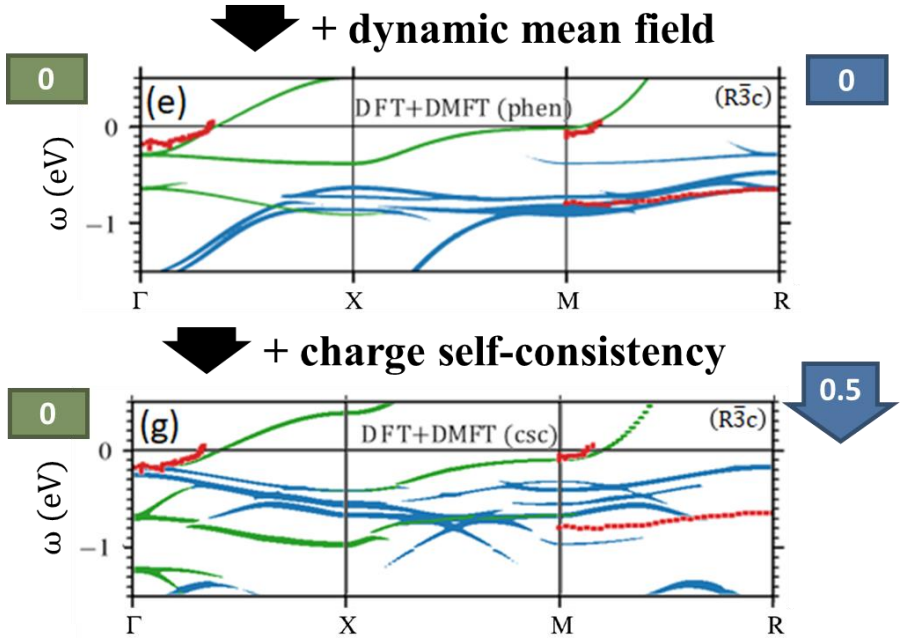
degrees of freedom





$LaNiO_3$



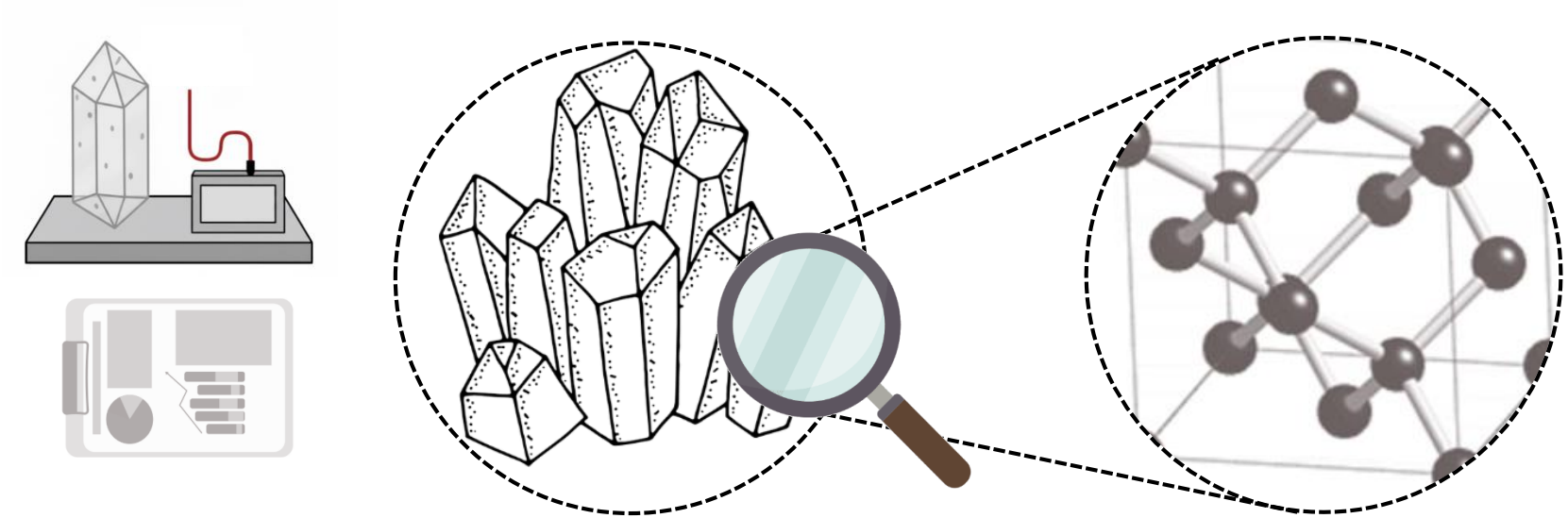


DMFT + rhombohedral (2 Ni-d + 6 O-p)

✓ subtle lattice distortions

✓ strong electronic correlations

⇒ beyond DMFT approximation: non-local correlation



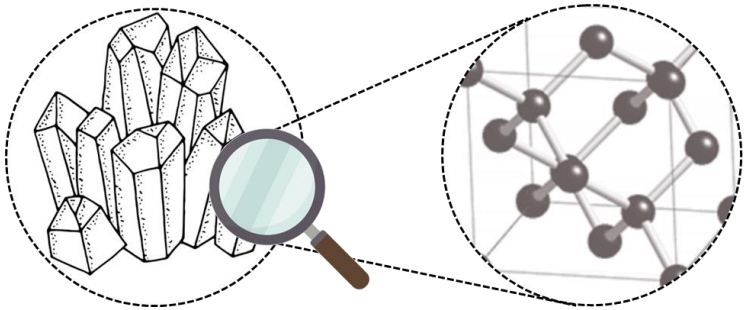
Macroscopic

Microscopic

materials like metals, semiconductors, …

how electrons behave within solids …



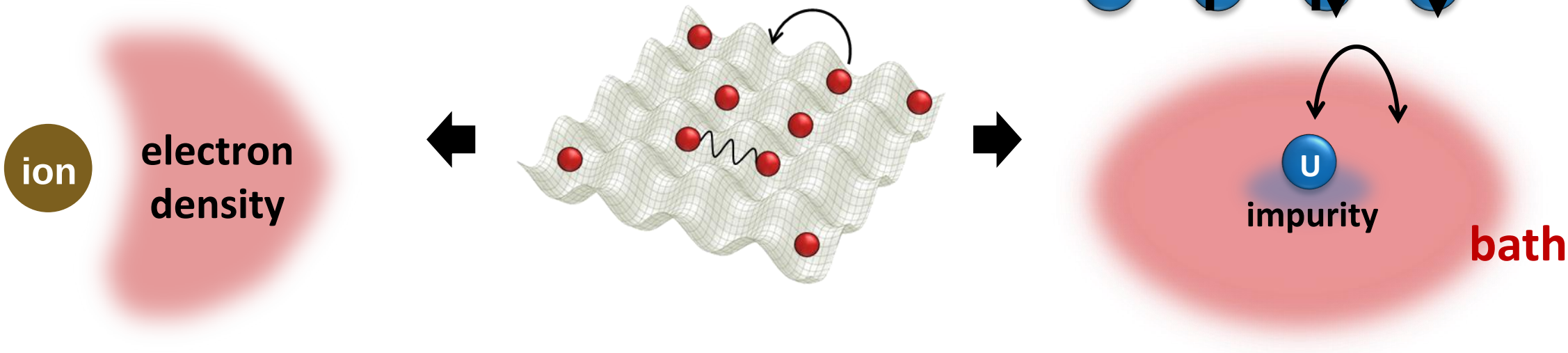


eigenvalue problem

$$H\psi = E\psi$$

Configuration Interaction

$$\psi = \sum_{i=1}^{601080390} c_i \phi_i$$



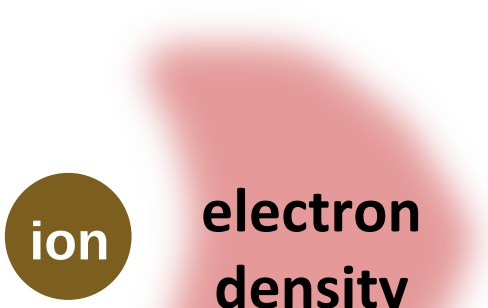
Density Functional Theory

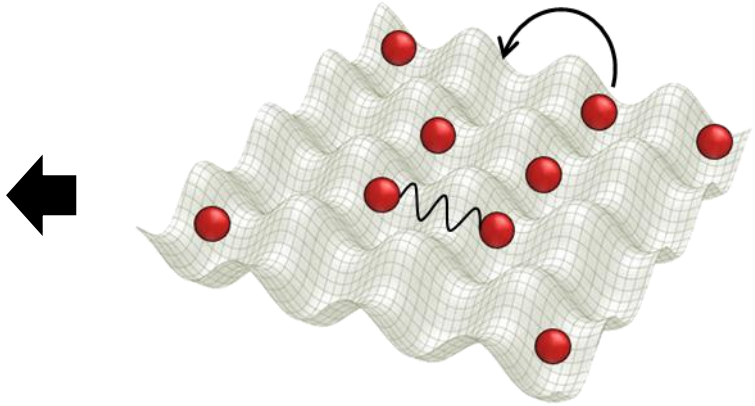
Dynamical Mean Field Theory

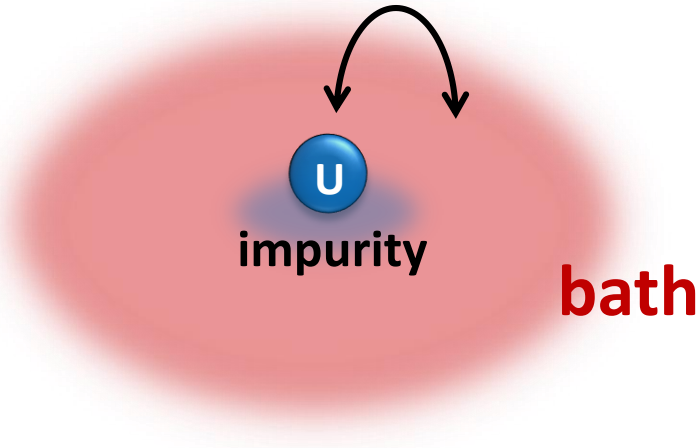












Density **F**unctional **T**heory

Dynamical **M**ean **F**ield **T**heory

QuBear 量子熊

量子熊 × 泛科學 第二季 ▶ 全部播放



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量子超級英雄
47-48



